An Intelligent Energy Management Model Integrating Renewable Energy and Smart Grids for Optimal Power Distribution

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***Abstract*— The inadequacy of traditional power networks and domestic energy management systems to efficiently optimize energy use results in excessive costs and increasing carbon emissions. Real-time dynamic power distribution management is impossible with these systems, particularly when dealing with several renewable energy sources (RES). This research suggests an intelligent energy management (IEM) model that optimizes power distribution by combining smart grids, battery storage systems (BSS), and RES. The research aim is to reduce Power prices, peak-to-average ratio (PAR), and carbon (CO_2) emissions while ensuring user comfort and system stability. The IEM model employs energy scheduling and management strategies that include heuristic optimization methods, like Dynamic Walrus-Gazelle Optimization (DWGO), a hybrid optimization algorithm combining the Enhanced Walrus Optimization (EWOA) and Gazelle Optimization (GOA). The model optimizes energy consumption and storage based on real-time electricity costs, renewable energy availability, and load demand. The experiment uses Python 3.7.0 for implementation. According to numerical evaluations, power prices, PAR, and CO_2 emissions have been significantly reduced respectively, in contrast to unscheduled. The suggested DWGO improves energy distribution while lowering costs and emissions and increasing system efficiency, making it a viable solution for reliable IEM models in smart buildings.**

Keywords*— *Smart grids, renewable energy sources (RES), intelligent energy management (IEM), energy distribution.

I. Introduction

Increasing world demand for power and the requirement to be sustainable have led to the integration of renewable energy resources (RES) into contemporary grid systems. However, the intrinsic intermittency of RES like solar and wind power is a reason for the problems of stable grids and the maximum delivery of power [1]. To combat the problems, smart grids developed to be more sophisticated with real-time sensing, predictive capability, and remote energy management [2]. A reliable energy management system utilizing renewable energy and smart grids guarantees optimum power supply, efficiency, and continuity [3]. The use of modern technologies like the Internet of Things (IoT), artificial intelligence (AI), and demand-side management makes the use of smart grids efficient in terms of dynamic load balancing and adaptive energy distribution. Energy storage facilities also facilitate the achievement of this integration by guaranteeing lower supply and grid vulnerability fluctuations [4]. It is predicated on the gathering of real-time data, forecasting energy supply and demand, and using optimization algorithms to achieve the most efficient energy flow balance. It also includes demand response strategies where customers consumption is adjusted based on the grid condition and dynamic price policies. The smart grid thereby enables more renewable energy to be used effectively while mitigating energy waste, lowering operational costs, and enhancing sustainability [5]. The constraints include high installation costs, cybersecurity threats, intermittent renewable energy feed, intricate infrastructure demand, data privacy issues, and the requirement for sophisticated forecasting and real-time control to enable optimal grid balance.

The objective of the DWGO is to provide optimal power supply by harnessing smart grids, energy storage units, and renewable energy systems, decreasing power price, PAR, and CO_2 emission, while ensuring system stability and user convenience.

II. Related Works

Through the integration of renewable energy sources and the implementation of a real-time price-based demand response program, the research [6] seeks to increase the efficiency of the smart grid. It suggests an intelligent framework for demand-side management that uses HGAC optimization algorithms and artificial neural networks. To improve smart grid sustainability, the experiment [7] aims to

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create an AI-based energy management system for SPV microgrids. Utilized a modified Grey Wolf Optimizer for optimal energy dispatch in grid-connected systems, which has been proven under various weather situations. Cost reductions of 23.34% and 45.55% were achieved on clear and overcast days, respectively. By combining battery storage and renewable energy sources, the evaluation [8] seeks to enhance energy management by lowering expenses, carbon emissions, and peak-to-average ratios. It suggests a controller for energy storage and load scheduling that makes use of heuristic algorithms. The objective of the evaluation [9] was to use Lyapunov optimization (LOT) to optimize energy in real-time in a smart house with an emphasis on thermal comfort and energy cost. The findings demonstrate a 20.15% decrease in energy expenses while satisfying user needs. For smart grid energy management optimization, the research [10] created a hybrid genetic algorithm (GA) and enhanced differential evolution (EDE) called EDGE. Use EDGE in an Energy Management Controller (EMC) to react to demand response (DR) signals and plan residential loads. EDGE was superior to other methods in terms of reducing carbon emissions, energy expenses, the PAR, and user comfort.

III. METHODOLOGY

The approach is to design an IEM model based on Dynamic Walrus-Gazelle Optimization (DWGO) that optimizes power allocation, energy scheduling, and storage concerning real-time cost, renewable supply, and load demand. Fig. 1 depicts the proposed system flow.

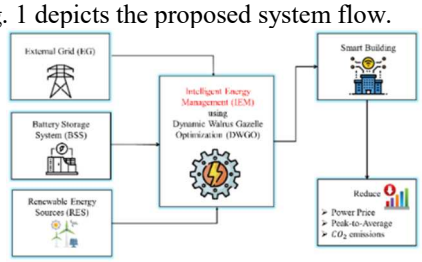

Fig. 1. Flow model of proposed system

A. Dynamic Walrus-Gazelle Optimization (DWGO)

DWGO is a hybrid heuristic method that combines the DWOA and GOA to efficiently balance energy. DWGO utilizes walrus searching modes for global exploration and gazelle-evading mechanisms for exploitation. DWGO dynamically regulates energy scheduling in smart grids through the maximum use of renewable energy, battery, and grid dependency. This guarantees lower power prices, lower PAR, and lower CO_2 emissions, making it perfect for smart energy management systems.

a. Dynamic Walrus Optimization Algorithm (DWOA)

The DWOA enhances the conventional optimization based on Lévy flight, which is a stochastic movement strategy with an abrupt change in direction and extended travel distances. The approach enhances global search, avoids premature convergence, and enhances solution precision in energy optimization problems. For managing the energy of the smart grid, DWO performs position updates using Lévy flight to better manage energy resources, optimize renewable energy dispatch, and improve system stability. Dynamic adjustment of energy scheduling lowers electricity bills, PAR, and CO_2 emissions. Lévy flight in DWO is expressed in (1) as follows:

$$LF = 0.01 \times \frac{v \times \sigma}{|u|^{\frac{1}{2}}}, \sigma = \left(\frac{\Gamma(1+\Upsilon) \times \sin\left(\frac{\pi\Upsilon}{2}\right)}{\Gamma\left(\frac{1+\Upsilon}{2}\right) \times \Upsilon \times 2\left(\frac{\Upsilon-1}{2}\right)} \right)^{\frac{1}{\Upsilon}} \quad (1)$$

Where v and u are random variables in (0, 1), integrating randomness into the search process. Using Lévy flights, the EWO algorithm improves the energy management model to offer maximum power distribution, efficient use of the grid, and renewable utilization of energy.

b. Gazelle Optimization Algorithm (GOA)

GOA is also utilized in the IEM model for power supply scheduling within smart grids that consist of a combination of RES and BSS. GOA relies on natural gazelle foraging and evasive behavior to perform energy scheduling, lower electricity costs, and ensure grid stability. This model simulates the movement of the gazelles in the solution space in different energy distribution situations. The most powerful gazelles (Elite solutions) control the power optimization distribution for maximizing the energy content of RES, BSS, and grid supply. The following (2) is the elite matrix for the optimal energy allocation.

$$Elite = \begin{bmatrix} w'_{1,1} & w'_{1,2} & \cdots & w'_{1,c} \\ w'_{2,1} & w'_{2,2} & \cdots & w'_{2,c} \\ \vdots & \vdots & \ddots & \vdots \\ w'_{m,1} & w'_{m,2} & \cdots & w'_{m,c} \end{bmatrix} \quad (2)$$

The modified energy distribution is represented by the following (3) as follows.

$$gazellet + 1 = gazellet + t \cdot Q \cdot QA \cdot (elitet - QA \cdot gazellet) \quad (3)$$

Where QA stands for Brownian motion in adaptive energy allocation, Q is a random variable, and t is the adjustment factor. Even though the grid experiences fluctuations (such as when there are predators), the algorithm automatically adjusts energy scheduling with Lévy flight motion in (4).

$$gazeller + 1 = gazellet + T \cdot \mu \cdot QK \cdot (Elitet - QK \cdot gazellet) \quad (4)$$

Where the grid response rate is T, and QK depends on real-time load changes. The adaptive demand response control starts with cumulative grid supply and demand variations, and these are presented as follows in (5).

$$DE = \left(1 - \frac{s}{S}\right)^{\frac{2s}{S}} \quad (5)$$

Providing grid stability with reduced peak demand and carbon emissions. With the incorporation of GOA in the IEM model, there is real-time energy optimization, demand response, and load balancing. The algorithm significantly reduces electricity bills, peak demand, and CO_2 emissions, and hence it is a strong solution for smart grid energy management.

B. Power bills

The real energy cost per time interval was determined on a Real-Time Pricing rate. Energy cost per hour is determined by the following (6).

$$\zeta(s) = \left(\Gamma_u(s) + \Gamma_E(s)\right) \times FO(s) \quad (6)$$

Where Γ_u^s and Γ_{tg}^s stand for the energy usage of stationary and movable appliances, respectively. The cost of power per day may be determined as follows in (7). Equations (8-9) represent the supplied power from BSS and RES. Equation (10) provides the entire quantity of power consumed.

$$\zeta(S) = \sum_{s=1}^{S}\left(\Gamma_u^s(S) + \Gamma_E^s(S)\right) \times FO(s) \quad (7)$$

$$\emptyset(s) = \left(\Gamma_{mt} + \Gamma_{tg}(s)\right) - \left(F(s) + BSS.\alpha_{bss}(s)\right). \quad (8)$$

$$\emptyset(s) = \begin{cases} \emptyset(s), & if\ \emptyset(s) \geq 0 \\ 0, & moreover \end{cases} \quad (9)$$

$$\delta(S) = \sum_{s=1}^{S}\left(\emptyset(s) \times FO(s)\right) \quad (10)$$

PAR: To reduce PAR, the IEM Model tries to optimize the use of energy. This is by RES, BSS, and EG. PAR for M consumers can be calculated using (11) as follows:

$$\delta(S) = \frac{\max\left(F_{total}(s)\right)}{\frac{1}{S}\sum_{m=1}^{M}\left(\sum_{s=1}^{M} F_{total}(s,m)\right)} \quad (11)$$

By lowering PAR, the technology gives a constant power supply, decreased peak demand, and improved energy efficiency.

CO_2 Emission: The carbon produced by using power is displayed in (12). Where FO_{avg}, η, γ, and n stand for the average monthly energy bill, price per kWh of \$0.20, electricity emission factor equivalent of 1.37, and calendar month, respectively.

$$\theta(s) = \frac{FO_{avg}}{\eta \times \gamma \times n} \quad (12)$$

The approach facilitates better energy usage in smart networks, improves sustainability, and reduces CO_2 emissions through energy schedule optimization.

C. System Model

The proposed system of the IEM model combines BSS, RES, and EG to maximize power distribution. It utilizes *DWGO* for real-time scheduling of energy.

EG: The smart building purchases energy from the EG at an established bill, as detailed in the subsequent sentence. The structure receives its energy from its external electricity system. Let $F_{fw}(s)$ show how much electricity was taken from the power grid at a certain time s, (13).

$$0 \leq F_{fw}(s) \leq F_{fw,max} \quad (13)$$

RES: RES with wind power refers to collect wind energy using turbines to create electricity. Wind turbines generate electricity based on wind speed, which is influenced by environmental factors and path of the wind. The quantity of energy produced by a wind turbine is estimated using the following (14).

$$O_{xc}(s) = \frac{1}{2} * \rho * B_{sc} * U_t(s)^3 \quad (14)$$

The characters ρ and B_{sc} represent the turbine blade's surface and *wind* volume, correspondingly, while $U_t(s)$ represents the speed of the air in m/s.

BSS: BSS accommodates excess RE to be used as a recovery in high-demand times such that the charging and discharging processes are optimized. Stored energy at time s is expressed as the following (15).

$$BS(s) = BS(s-1) + l.\delta^{BS}.FO^{dg}(s) - \frac{l.FO^{cdg}(s)}{\delta^{BS}} \quad (15)$$

Where $BS(s)$ is stored energy, l is the duration of a time slot, δ^{BS} is the BSS efficiency, $FO^{dg}(s)$ is charging power, and $FO^{cdg}(s)$ is discharging power. To keep the battery healthy and avoid overcharging or deep discharge, the system using (16-18) as follows:

$$FO^{dg}(s) \leq FO_{VA}^{dg} \quad (16)$$

$$FO^{cdg}(s) \leq FO_{KA}^{cdg} \quad (17)$$

$$BSS(s) \leq FT_{VA}^{dg} \quad (18)$$

This provides an uninterrupted power supply, improved energy utilization, and longer battery life in intelligent grids.

II. Experimental Results And Discussion

The experiment and effects of the DWGO were demonstrated. Windows 10 64-bit with a 2.80 GHz Intel Core i7-6700T CPU and 16 GB of RAM was used for the experiment. Three instances were examined: case 1 (EG): Depends entirely on the external grid for power. Case 2 (EG + RES) distributes electricity using both renewable energy sources and external grid power. Case 3 (EG + RES + BSS): Combines external grid, renewable energy, and battery storage for effective energy management. To assess whether the suggested scheduled algorithm and existing scheduled algorithms perform. The performance is compared based on a 24-hour evaluation.

Case1(EG only): IEM employs EG Electricity for scheduled and *unscheduled* (Unsc) loads. It addresses reducing the power price, PAR, and CO_2. BFO, and HGPDO significantly lower power bill prices (52.54%, and 58.69%, respectively). However, the proposed scheduling technique achieves optimal price reduction. Table I displays the average 24-hour price of traditional methods compared to the proposed model. Fig. 2 illustrates the case 1 power price (Wh) contrast. Unsc Power price ranges from an extreme of 1091 cents in hour 9 to a lowest of 165.81 cents in 16 hours. In case 1, DWGO decreases the power price to 75.6 in 16 to 22 hours.

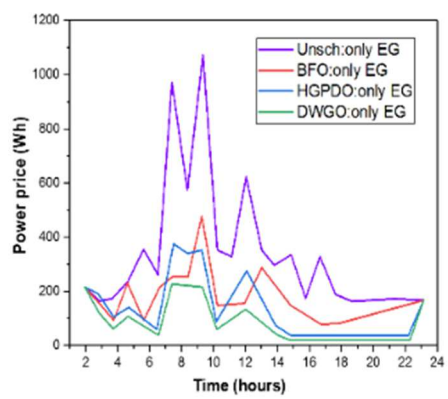

Fig. 2. Case 1 Comparison of Power price (Wh)

TABLE I. CASE 1 CONTRAST OF POWER PRICE

Algorithms	Normal price (cents)	Variance (normal cents)	Decrease (%)
Unsc	337.34	--	--
BFO [8]	160.07	177.2	52.54%
HGPDO [8]	138.82	198.52	58.54%
DWGO [proposed]	125.02	212.32	62.93%

Table II and Fig. 3 show Case 1 PAR comparison, the $unsc$ method has the highest PAR (3.302), which implies inefficient load balancing. BFO, HGPDO, and the proposed DWGO significantly decrease PAR with the lowest value (0.753) for DWGO a 77.19% decrease, which reflects its potential for *reducing* energy distribution waste and peak load. Table III and Fig. 3 show CO_2 emission comparison, the desired CO_2 content (D) is 760ppm. The unsc method increases CO_2 emissions (1014 ppm). Emissions are minimized by optimization algorithms, DWGO minimizes the minimum (64.5 ppm) by 91.44%. This identifies DWGO's better result in reducing environmental load via effective energy management.

TABLE II. CASE 1 COMPARISON OF PAR.

Algorithms	Overall PAR	Variance (PAR)	Decrease (%)
Unsc	3.302	--	--
BFO [8]	2.654	0.648	19.62%
HGPDO [8]	1.559	1.743	52.78%
DWGO [proposed]	0.753	2.549	77.19%

TABLE III. CASE 1 CONTRAST OF CO_2 EMISSION

Algorithms	CO_2 emission(ppm)	Variance(ppm)	Decrease(%)
CO_2 desired	760	--	(Increased)
Unsc	1014.0	-254	-33.42%
BFO [8]	534.8	225.2	29.63%
HGPDO [8]	72.4	687.6	72.40%
DWGO [proposed]	64.5	695.5	91.44%

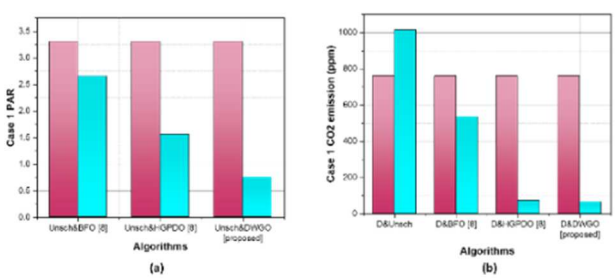

Fig. 3. case 1 (a) PAR and (b) CO_2 emission

Case 2 (EG and RES combination): The IEM uses both external grid power and RES for scheduled and unsc tasks. BFO, HGPDO, and DWGO significantly lower power prices by 25.27%, 47.55%, and 51.20%, respectively. In case 2, the suggested DWGO scheduling method achieves the best cost-reduction results. Table IV compares the average cost of all methods over 24 hours, including the suggested approach in example 2. The electricity price (Wh) contrast in case 1 is shown in Fig. 4. The $Unsc$ price of Power ranges from a peak of 722.35 cents in hour 9 to a minimum of 109.10 cents in 16 hours. In casa 2, the power price was decreased by DWGO by 50.3 during 22 hours.

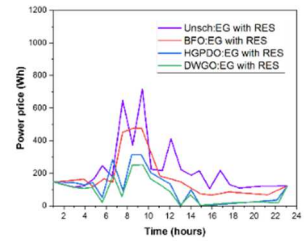

Fig. 4. Case 2 Comparison of Power price (Wh)

TABLE IV. CASE 2 POWER PRICE COMPARISON.

Algorithms	Normal price (cents)	Variance (normal cents)	Decrease (%)
Unsc	225.91	--	--
BFO [8]	168.82	57.09	25.27%
HGPDO [8]	118.48	107.43	47.55%
DWGO [proposed]	110.24	115.67	51.20%

Table V and Fig. 5 (a) show case 2 PAR comparison, the unsc strategy possesses the largest PAR (2.985), which represents the worst load balancing. The algorithms perform better with BFO (26.80%), HGPDO (45.02%), and DWGO possesses the lowest PAR (0.851), a decrease of 71.49%. This suggests DWGO's higher efficiency in optimizing energy distribution and minimizing peak load demand. Table VI and Fig. 5 (b) depict the CO_2 emissions for case 2. The desired carbon content (D) is 760ppm. The unsc approach is connected to a higher emission of CO_2 (1003 ppm, 31.97% increase). Optimization techniques significantly minimize emissions, DWGO having the lowest (44.2 ppm) a 94.18% decrease, reflecting DWGO better capability to promote sustainability through effective energy management.

TABLE V. CASE 2 PAR COMPARISON.

Algorithms	Overall PAR	Variance (PAR)	Reduction (%)
Unsc	2.985	--	--
BFO [8]	2.185	0.800	26.80%
HGPDO [8]	1.651	1.344	45.02%
DWGO [proposed]	0.851	2.134	71.49%

TABLE VI. CASE 2 CONTRAST OF CO_2 EMISSION

Algorithms	CO_2 emission (ppm)	Variance (ppm)	Decrease (%)
CO_2 desired	760	--	--
Unsc	1003	-243	-31.97% (Increased)
BFO [8]	594	166	21.84%
HGPDO [8]	53.4	706.6	92.90%
DWGO [proposed]	44.2	715.8	94.18%

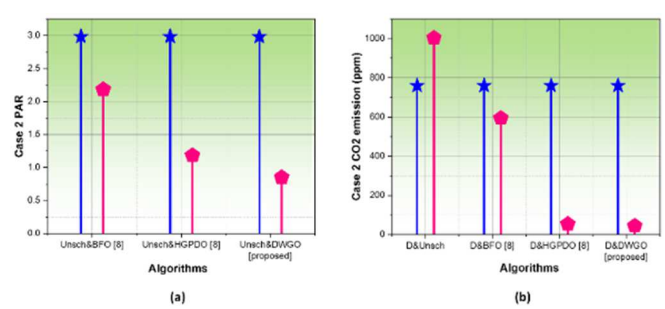

Fig. 5. Case 2 (a) PAR and (b) CO_2 emission

Case 3 (EG combined with BSS and RES): The IEM in this case uses EG power for both scheduled and unsc tasks, in addition to RES and BSS. The total price of the power bill shows that BFO, HGPDO, and DWGO respectively, lower the cost by 29.72%, 33.66%, and 43.72%. However, in Case 3, the suggested scheduling approach, which evenly distributes demand between low and high peak hours, achieves the optimum price reduction. Table VII and Fig. 6. shows the average cost per twenty-four hours for all techniques and the proposed solution in Case 3. The unsc of power price rises during time hour 9, with the biggest electricity cost being 711.82 cents. Lowe's price was 102.88 cents in slotted hour sixteen. In case 3, DWGO power prices decreased by 45.2 in 23 hours.

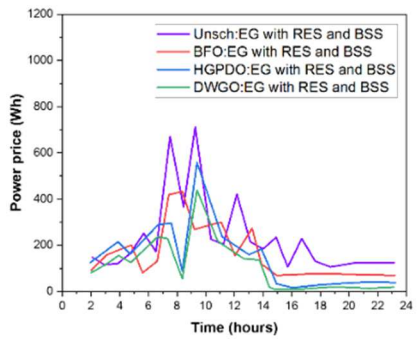

Fig. 6. Case 3 Comparison of Power Price (Wh)

TABLE VII. CASE 3 COMPARISON OF POWER PRICE

Algorithms	Normal price (cents)	Variance (normal cents)	Decrease (%)
Unsc	222.51	--	
BFO [8]	156.38	66.13	29.72%
HGPDO [8]	147.61	74.9	33.66%
DWGO [proposed]	125.21	97.3	43.72%

Table VIII and Fig. 7(a) show case 3 PAR comparison, the unsc method yields the maximum PAR (2.537), a measure of worst load balancing. The performance is enhanced with the assistance of optimization algorithms, wherein BFO (31.92%), HGPDO (54.35%), and DWGO yield the minimum PAR (0.821) an enhancement of 67.63%. This shows DWGO's better capacity to optimize energy supply and peak loads effectively. Table IX and Fig. 7 (b) depict the CO_2 emissions for case 3. The Unsc method raises CO_2 emissions (968.3 ppm, 27.40% increase). Optimization methods reduce emissions considerably, DWGO reducing the lowest amount (52.3 ppm) by 93.11%, showing DWGO's efficiency in reducing environmental load through optimized use of renewable energy.

TABLE VIII. CASE 3 CONTRAST OF PAR.

Algorithms	Overall *PAR*	Variance (*PAR*)	Reduction (%)
Unsc	2.537	--	--
BFO [8]	1.727	0.81	31.92%
HGPDO [8]	1.158	1.379	54.35%
DWGO [proposed]	0.821	1.716	67.63%

TABLE IX. CASE 3 CONTRAST OF CO_2 EMISSION

Algorithms	CO_2 emission (ppm)	Variance (ppm)	Decrease (%)
CO_2 desired	760	--	--
Unsc	968.3	−208.30	−27.40% (Increased)
BFO [8]	567.5	193.00	25.39%
HGPDO [8]	63.5	696.5	91.64%
DWGO [proposed]	52.3	707.7	93.11%

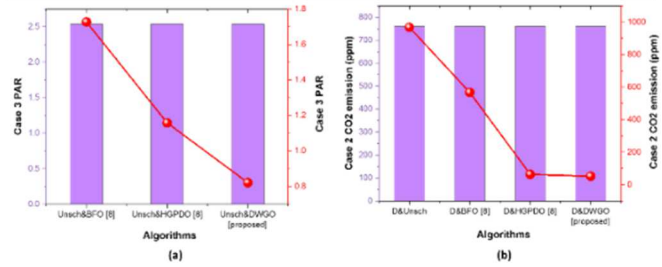

Fig. 7. Case 3 (a) PAR and (b) CO_2 emission

IV. Conclusion

The evaluation was intended to overcome the challenges of traditional home energy management systems and power grids, unable to optimize energy utilization, resulting in high carbon emissions and prices. It assessed an IEM system comprising smart grids, BSS, and RES for minimizing power distribution. The model utilized heuristic optimization methods such as *DWGO*. Numerical results revealed significant IEM decreases in power prices, PAR, and CO2 emissions in case 1, by 62.93%, 77.19%, and 91.44%, in case 2, 51.20%, 71.49%, and 94.18%, in case 3, 43.72%, 67.63%, and 93.11%, respectively, as compared to the unsc method. The DWGO-based IEM model proved to be capable of increasing the distribution of energy, minimizing expenses, lowering emissions, and increasing overall system efficiency, and thus it has effective potential for use in smart building energy management. This approach has limitations, including high startup costs, integration complexity, and reliance on weather-dependent renewable sources. Future plans include developing AI-driven predictive analytics, implementing blockchain to improve grid resilience and integrating energy storage technologies to provide sustainable and reliable power distribution in changing energy landscapes.

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